

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	12 July 2026
Team ID	SWTID-2026-6951
Project Name	Visualization Tool for Electric Vehicle Charge and Range Analysis
Maximum Marks	4 Marks

Technical Architecture:

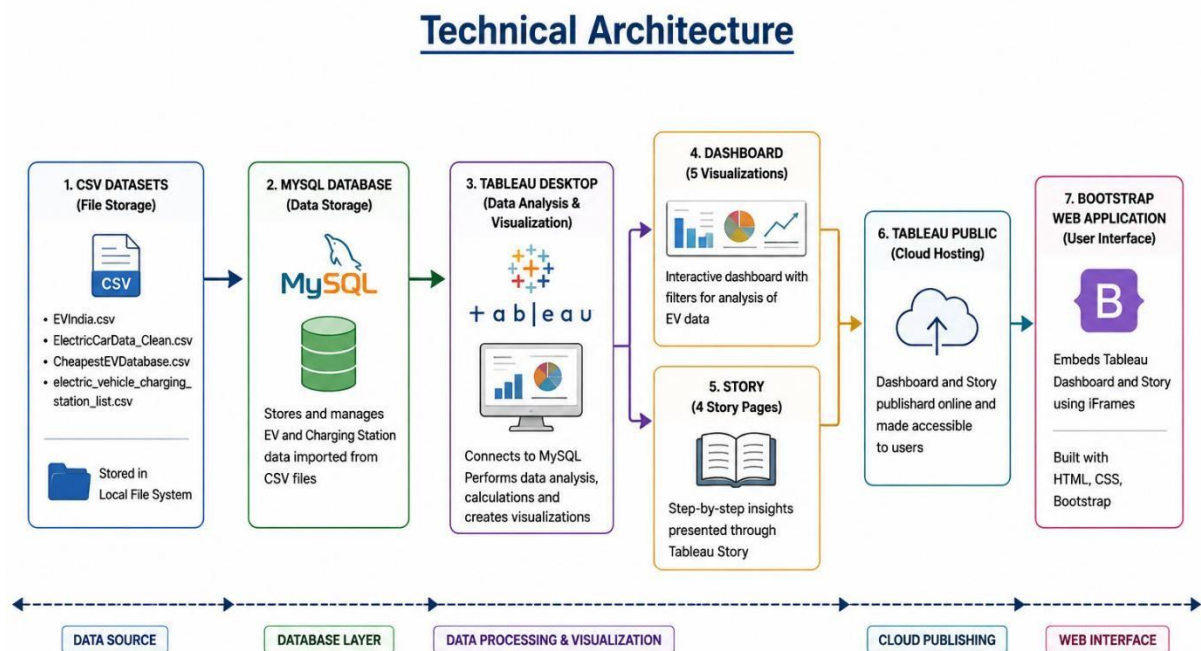


Table-1 : Components & Technologies:

1.	User Interface	Web interface for viewing dashboard and story	HTML, CSS, Bootstrap
2.	Application Logic-1	Data filtering, sorting and visualization	Tableau Desktop
3.	Application Logic-2	Interactive Dashboard and Story	Tableau Dashboard & Story
4.	Database	Stores EV datasets	MySQL

5.	Cloud Database	Not Applicable	-
6.	File Storage	Stores CSV datasets	Local File System
7.	External API-1	Not Used	-
8.	External API-2	Not Used	-.
9.	Machine Learning Model	Not Used	-
10.	Infrastructure	Dashboard published online and embedded into website	Tableau Public + Local System

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Responsive web application developed using Bootstrap	Bootstrap 5
2.	Security Implementations	Dashboard is publicly accessible in read-only mode	Tableau Public
3.	Scalable Architecture	New datasets and visualizations can be added easily	Tableau + MySQL
4.	Availability	Dashboard and Story are available online through Tableau Public	Tableau Public
5.	Performance	Optimized dashboard with interactive filters for faster visualization	Tableau + MySQL

References:

- Tableau Public Documentation
- MySQL Documentation
- Bootstrap Documentation